

Fernando d'Ávila Lins Bezerra Cavalcanti Filho

Software Engineering | Full Stack Development | Artificial Intelligence

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GitHub | LinkedIn | Portfolio

Professional Summary

Computer Science undergraduate at the Federal University of Paraíba with hands-on experience in Full Stack development, backend systems, REST APIs, system integrations, workflow automation, and Artificial Intelligence. Experienced in building applications with Python, TypeScript, JavaScript, Node.js, Next.js, React, and PostgreSQL. Academic and practical experience with Machine Learning, NLP, LLMs, RAG, OCR, and model evaluation. Analytical, collaborative, and problem-solving-oriented, with the ability to learn new technologies quickly and translate business needs into software solutions.

Technical Skills

Programming Languages: Python, TypeScript, JavaScript, Java, C, and C++.

Frontend: React, Next.js, Tailwind CSS, HTML, and CSS.

Backend and APIs: Node.js, REST APIs, HTTP, JSON, system integrations, and layered architecture.

Data: PostgreSQL, SQL, NoSQL databases, data modeling, Pandas, and NumPy.

Artificial Intelligence: Machine Learning, NLP, LLMs, RAG, OCR, prompt engineering, PyTorch, Scikit-learn, and LangChain.

Engineering and Tools: Git, GitHub, Docker, Linux, Postman, Scrum, and Agile methodologies.

Professional Experience

Full Stack Development Intern

Acutis Tecnologia

May 2024 – Present | Next.js, Node.js, TypeScript, SQL, NoSQL, Git, and Scrum

- Developed a project management dashboard that improved information visibility and facilitated the monitoring of internal processes.
- Designed and implemented an OCR pipeline for automated invoice data extraction, reducing manual work and data-entry errors.
- Developed a flashcard application for learning review and reinforcement, working with business rules, user interfaces, and data persistence.
- Built and consumed REST APIs for communication between frontend, backend, databases, and external services.
- Participated in requirements gathering, feature implementation, technical testing, bug fixing, and evolutionary maintenance of applications.
- Collaborated on the LOTEPE government project in an Agile environment, using Git for version control and team collaboration.

Software Development Intern

OpenDoor App

Jun. 2025 – Present | Next.js, Node.js, TypeScript, PostgreSQL, REST APIs, and Git

- Contribute to the development of a peer-to-peer home-swapping accommodation platform.
- Implement Full Stack features, business rules, and integrations between the frontend, backend, and database.
- Participate in data modeling, API endpoint creation and evolution, and improvements to the application architecture.
- Collaborate on bug fixing, performance improvements, and continuous product development, considering scalability and user experience.

Education and Technology Intern

Colégio e Curso Evolução

Jan. 2024 – Jan. 2025 | Python, Google Colab, and Data Optimization

- Developed a system to support teacher allocation across middle school and high school classes.
- Automated an administrative process previously performed manually, reducing schedule conflicts and improving workload distribution.
- Gathered requirements from institutional teams and translated educational process rules into a computational solution.
- Supported educational innovation initiatives and the adoption of technology in internal institutional processes.

Extension, Research, and Academic Leadership

Software Developer – Structuring Businesses Extension Program

UFPB

Aug. 2022 – Jul. 2023 | TypeScript, JavaScript, and Lean Startup

- Delivered, together with other university students, activities on Lean Startup, hypothesis validation, business modeling, and product development.
- Collaborated with multidisciplinary teams to support entrepreneurs and structure business ideas.

- Participated in requirements gathering and the development of a web application for managing vaquejada events using TypeScript and JavaScript.
- Applied MVP and needs-validation concepts to transform a business problem into a functional software solution.

Member – NLP Division Rotation **TAIL – UFPB Artificial Intelligence Academic League**
Python, LLMs, NLP, Prompt Engineering, and Model Evaluation

- Participated in the project “Evaluating LLM Performance for Product Description Normalization in Brazilian Electronic Invoices”.
- Collaborated on the evaluation of language models applied to the standardization of unstructured product descriptions.
- Worked on prompt definition, output comparison, consistency analysis, and the development of evaluation criteria.
- Developed a sentiment analysis prototype aimed at improving the expressiveness of virtual characters in the VLibras project.

Teaching Assistant – Vector Calculus and Analytical Geometry **UFPB**
May 2023 – Dec. 2023

- Conducted in-person and remote tutoring sessions, helping students understand concepts and solve mathematical problems in a structured way.
- Prepared supporting materials and explained technical content clearly to students with different levels of prior knowledge.
- Received the **2023 Teaching Initiation Award – 1st Place**, granted by UFPB in recognition of excellence in academic tutoring.

Selected Projects

Multiple Sclerosis Lesion Detection
Python, PyTorch, Scikit-learn, and Medical Imaging

- Develops Machine Learning models for the identification and segmentation of multiple sclerosis lesions in MRI scans.
- Works with image preprocessing, model training, evaluation, and collaboration with LABEN researchers.

Academic Assistant with LLM and RAG
Python, LangChain, LLMs, RAG, and Vector Databases

- Developed a conversational assistant to answer academic and administrative questions using official UFPB documents as its knowledge source.
- Implemented information retrieval, context construction, and answer generation based on retrieved content.

Interactive Quiz Simulator
Java, Object-Oriented Programming, and Concurrency

- Implemented a quiz application using Java, object-oriented programming concepts, and concurrent programming to simulate interactive sessions.

Education

Federal University of Paraíba – UFPB **João Pessoa – PB, Brazil**
Bachelor’s Degree in Computer Science | 2022 – 2027
GPA: 8.38/10.
Relevant Coursework: Data Structures, Design and Analysis of Algorithms, Object-Oriented Programming, Software Engineering, Operating Systems, Computer Networks, and Database Systems.

Additional Activities

TRIL – Software Development Laboratory: participation in software development activities and application projects.
LABEN – Aging and Neuroscience Research Laboratory: collaboration on Artificial Intelligence research applied to medical imaging.
Formula E-Motion – UFPB: participation in embedded systems, telemetry, CAN bus communication, and circuit development using KiCad.

Languages

Portuguese: Native | **English:** Advanced | **Spanish:** Intermediate